A062—SST Microgravity Force-Gradient Falsifier

v0.3.0: Euler Pressure Monopole versus Conditional Swirl-Clock EFT

SST-Workbench / source-pinned technical protocol

13 October 2026

1 Scientific scope and identification

Three hypotheses are independently recorded. H0 is the localized incompressible Euler pressure source; H1 is a *postulated* massless scalar mediator with a normalized nonzero charge; H2 asks whether H0 and authentic E011 trefoil dynamics generate the H1 source and universal physical coupling. The H1 result is a conditional mathematical prediction, not an SST derivation of gravity. E011 `STATIC_READY` does not imply dynamically stationary knots.

Automatic run provenance

Field	Recorded value
Run ID	20261013T100405915583Z-70837109
Run phase	BLIND
Protocol bundle SHA-256	76b9fa0bf0d2a7fccab5729ebcc3ef4239ae87e498c947f0ea249c9e3aa7fb41
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Output manifest SHA-256	e2f40147cc5df133402acf6ff7f1871ab09f2eecda3199cf3c9cb84db5c12570

Gate ledger

Gate	Status	Question
G0	PASS	Has v1.0.8 verified the frozen instance protocol?
G1	PASS	Are both raw E011 v0.3.0 geometry hashes independently verified?
G2	PASS	Do Euler zero-monopole and conditional scalar EFT numerical controls converge?
G3	FAIL	Does Euler-only PKLSA generate a nonzero intrinsic Swirl-Clock monopole and universal
G4	UNRESOLVED	Are withheld microgravity raw-data replication measurements available?
G5	PASS	Does an independent compiled native C++ FP64 scalar kernel match Python?
G6	NOT_APPLICABLE	Did optional certified SYCL GPU processing execute?
G7	PASS	Do independent E011 geometry providers reproduce conditional EFT far-field exponents?
G8	UNRESOLVED	Is physical clock charge, mass equivalence, normalization and Euler-stress closure derived?
G9	DEFERRED	Was reveal separately authorized and verified?

Backend manifest


```
{
  "cpp_fp64": {
    "comparisons": [
      {
        "provider": "gilbert",
        "relative_error": 1.772222426323943e-15
      },
      {
        "provider": "knotplot",
```

```
        "relative_error": 9.542015328132388e-16
      }
    ],
    "compiler": "C:\\workspace\\Strawberry\\c\\bin\\g++.EXE",
    "executed": true,
    "max_relative_error": 1.772222426323943e-15,
    "status": "PASS"
  },
  "python_reference": {
    "actual_backend": "python-numpy-fp64",
    "authority": "REFERENCE",
    "precision": "float64"
  },
  "schema": "SST-BACKEND-MANIFEST-2",
  "scientific_boundary": "H1 inverse-square is a conditional result with explicitly postulated scalar c
  "sycl_gpu": {
    "authority": "SCREENING_ONLY",
    "requested": false
  }
}
```


Source manifest


```
{
  "manifest_sha256": "524a476a1561fa54d4e586e2b3c823cb035a6cfeda05628ccbbefd3f304c0b47",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [
    {
      "catalog_id": "E011",
      "evidence_class": "independent_source",
      "exists": true,
      "independence_family": "gilbert",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A062_SST_Microgravity
      "provenance_family": "gilbert_ideal_A004",
      "registry_hint": {
        "catalog_id": "E011",
        "hits": []
      },
      "role": "discovery",
      "sha256": "274a4ce111607a6295e280f3f8369c665e7de82ac33afcd4cb227eb61568d530",
      "source_id": "S_GILBERT",
      "version": "0.3.0"
    },
    {
      "catalog_id": "E011",
      "evidence_class": "independent_source",
      "exists": true,
      "independence_family": "knotplot",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A062_SST_Microgravity
      "provenance_family": "knotplot_relaxed_A001",
      "registry_hint": {
        "catalog_id": "E011",
        "hits": []
      },
      "role": "confirmation",
      "sha256": "8fea8a15ff3047dbbfb4aba05090b9a6cadc10eeedf97c22350925c764bc540e",
      "source_id": "S_KNOTPLOT",
      "version": "0.3.0"
    }
  ],
  "workbench_root": "C:\\workspace\\projects\\SST-Workbench"
```

}

2 Research question, hypotheses and isolation

Can the pressure-Poisson source of localized 3-D incompressible Euler create a long-range monopole directly? If not, does an independent clock scalar with nonzero charge yield the expected inverse square? Are the scalar charge and coupling determined by PKLSA dynamics without inserting G? The H0 and H2 null outcome is closure failure, not a conclusion about SST as a whole.

3 Euler-only pressure source (H0)

For $\nabla \cdot \mathbf{v} = 0$ with appropriate free-space decay,

$$\nabla^2 p = -\rho_f \, \partial_i v_j \partial_j v_i = -\rho_f \partial_i \partial_j (v_i v_j), \qquad \int_{\mathbb{R}^3} \nabla^2 p \, d^3x = 0. \tag{1}$$

The final integral vanishes only under the stated boundary and regularity conditions. This excludes a scalar *pressure-source* monopole in this restricted case, not all possible independent clock modes, nonlocal media or externally forced flows. The prior user-supplied A062-v0.2.0 Kelvin exchange shows a distinct R^{-4} observable; it is not automatically a material-particle force.

4 Conditional scalar field (H1)

Assume, without derivation from SST microphysics,

$$S[\tau] = \int d^3x \left(\frac{\kappa}{2} |\nabla \tau|^2 - \lambda \tau \rho_Q \right), \tag{2}$$

$$\kappa \nabla^2 \tau = -\lambda \rho_Q, \qquad \tau(\mathbf{r}) = \frac{\lambda}{4\pi\kappa} \int \frac{\rho_Q(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \, d^3 r'. \tag{3}$$

A *separate test-body coupling* is required to define an attraction. For the unit-test convention chosen here, $\kappa = \lambda = Q_A = Q_B = 1$ and $L = 1$; a two-knot force is evaluated using quadrature over authenticated E011 filament centerlines. All resulting forces are *dimensionless normalized EFT forces*, not Newtons. The normalized far field tends to $|F_z| \simeq 1/(4\pi R^2)$.

$$T_{ij}^{\text{cross}} = \kappa \left[(\partial_i \tau_A)(\partial_j \tau_B) + (\partial_i \tau_B)(\partial_j \tau_A) - \delta_{ij} \nabla \tau_A \cdot \nabla \tau_B \right]. \tag{4}$$

The cross-stress traction is integrated on a spherical surface enclosing B alone and compared with the scalar action-derived two-body force, with its explicit attraction/sign convention. A scalar Gauss flux $-\kappa \nabla \tau$ must not be described as mechanical momentum flux.

5 Unresolved source bridge (H2)

The provisional identification $\rho_Q \propto -\rho_f \partial_i \partial_j (v_i v_j)$ integrates to zero; it cannot explain a nonzero Newtonian monopole. To proceed one needs an independently derived SST clock scalar, an admissible nonzero source charge, a physical mass map and a dimensionful λ^2/κ. No such construction was supplied or inferred. Identifying $Q = M$ or fitting λ^2/κ to G would fail a first-principles closure test.

6 Numerics, source qualifications and backends

Raw E011-v0.3.0 sources: Gilbert `Ideal.txt.gz` record `3:1:1` and KnotPlot `coarse.final.vect`, both checked by SHA-256 against independent provider anchors. Geometries are resampled at $N = 48, 96, 192$, and each test source has length-normalized line weights summing to one. The scalar H1 pair force is evaluated at $R/L = 3, 5, 8, 12, 18, 27, 40$; fits use $R/L \geq 8$. Source direction and a rotated test configuration are recorded. Independent C++17 FP64 pair quadrature is compared against Python; a missing compiler leaves the native gate unresolved. An independent 3-D Euler pressure time integration is *not* claimed.

7 Gates, statistics and blindness

The unadjusted thresholds are scalar far-field $|n-2| < 0.05$, last resolution relative difference < 0.005, normalized provider-amplitude relative difference < 0.05, force reciprocity < 0.002, scalar cross-stress relative error < 0.07 and C++ parity $< 2 \times 10^{-10}$. A failed gate does not terminate the rest of the run. The published stopped-light experiment is already public and not a withheld confirmation series; this is an exploratory source-pinned computation, not a genuinely blinded experimental prediction. The framework's nonce, commitments, protocol freeze, run-ID separation and reveal restrictions are maintained.

8 Results and interpretation

The canonical framework v1.0.8 reports run-specific, nonoverwriting ZIP/PDF exports. Numerical H1 conditional success never promotes H2 to a pass. Source modes from the microscopic Euler vortices remain unproven. The physical microgravity experiment uses gram-scale bodies that cannot be identified with single E011 filaments without independent many-body mass closure.

9 Reproducibility

Within the bundled SST-Workbench structure, run `run_all.cmd FREEZE`, then `run_all.cmd CERTIFY`. The `FREEZE` launcher creates missing instance-private 32-byte nonce and HMAC key *only* before any frozen protocol or commitments exist; already present files are never changed. v1.0.8 is included as the original `LOCAL_VALIDATED_CANDIDATE` distribution and must not be represented as canonical-frozen without separate qualification.

References

[1] SST-47, *A First-Principles Origin of the Inverse-Square Law in Swirl–String Theory* (user-provided technical note).

[2] SST-48, *Emergent Inverse-Square Law from Hydrodynamic Dynamics* (user-provided technical note; corrected stress-divergence interpretation).

[3] SST-49, *A First-Principles Origin of the Inverse-Square Law* (user-provided technical note).

[4] SST-50, *Emergent Equivalence Principle from Relational Time and Connection Dynamics* (user-provided technical note).

[5] Y. Zhu et al., *Stopped-light-enhanced gravitational force sensing*, Nature Nanotechnology (2026), doi:10.1038/s41565-026-02298-8.